Ben Bubnick

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SKILLS

PROGRAMMING

Object Oriented:
C++ • C#.NET • PERL • Python
R • VB.NET

Other:
Batch • LaTeX • FORTRAN 90/95 • Regex

Query Scripting:
LINQ • SQL

Scientific:
Dataplot • IRAF • IDL • JMP
LabVIEW • Maple • Mathematica
MATLAB • PDL • SAS/GRAPH

Web application:
ASP.NET • CSS • HTML5 • JavaScript
JScript.NET • VBScript(IIS)

SCIENTIFIC

Computational Algorithms:
Christofides • Euler method
Demon Algorithm • Flow Networks
Finite Difference • Fisher–Yates
Gibbs Method • Monte Carlo
Newton's Method • Runge-Kutte

Interpolation:
Birkhoff • Hermite • Lagrange
Cubic , Monotone Cubic, Bicubic & Tricubic

LINKS

LinkedIn:// BenBubnick

Thesis: Massive Stellar Clusters

Published Data: JHK Photometry

EDUCATION

JOHNS HOPKINS UNIVERSITY

CS in Data Science
Concentration in R Programming
2015

UNIVERSITY OF CINCINNATI

MS in Physics
Concentration in Astrophysics
2010

OHIO UNIVERSITY

BA in Fine Arts
Concentration in Printmaking
2004

SUMMARY

- 9 years experience writing scientific software using various programming langaues
- Knowledge of modern object oriented programming languages: C#, C++, and Visual Basic
- Numerical methods for scientific research, development and exposition
- Scientific data processing and imaging
- Data simulation using Monte Carlo Methods such as Gibbs & Demon algorithms
- Team Lead on software development LOB, training new staff & co-ordinating daily allocation of work

WORK HISTORY

SOFTWARE ENGINEER | AmTrust Financial Services

2014 – Present | Cleveland, OH
Developed web forms for insurance industry using ISO standards and created multi-tier MVC components, Web APIs, and UIs using VB, ASP, Javascript, & HTML5

- Led development team on business integration of new aqcuisitions
- Implemented new unit testing method written in C#

COLOR SCIENTIST | PPG Industries

2012 – 2013 | Strongsville, OH
Developed software and radiative transfer models on complex paint mixtures to determine formulation methods and for spectroscopic profile analysis on color match samples.

- 40% marked improvement on color matching software algorithm using numerical integration methods
- Developed cross-instrument correlation software for statistical risk analysis projects using JMP
- Debugged optimization function for numerical solution to a version of the radiative transfer equation written in VB.NET and FORTRAN
- Wrote DLL wrapper for on-loan spectroscopic instrument using Visual C++ and VB.NET

ADJUNCT PROFESSOR | Miami University

2010 – 2011 | Hamilton, OH
Developed curricula in introductory physics & astronomy courses; shared time with Cincinnati State Community College

- Taught 3 courses: Astronomy, Introduction to Physics, and Physics & Society

RESEARCH

UC ASTROPHYSICS LAB | Graduate Research Assistant

2008 – 2010 | Cincinnati, OH
Worked with Prof Margaret Hanson to study Main Sequence stellar infrared spectroscopy. Performed quality control for the Cambridge Astronomical Survey Unit Variable Stars database (Publication) and controlled the SpeX spectroscopic tool as a visiting astronomer to the NASA Infrared Telescope Facility (Publication)